

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

9th March 2022

Subject SA4 – Pensions and Other Benefits

Time allowed: 3 Hours 30 Minutes (14.30 - 18.00 Hours)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

- 1. Please read the instructions to examinees sent along with hall ticket carefully and follow without exception.*
- 2. The answers are expected to be India Specific application for the syllabus and corresponding core reading. However, substantially the core reading material is still taken from material supplied by Actuarial Education Company which is meant for UK Fellowship examination. The core reading also contains some material which is India Specific, mostly the statutory laws and accounting standards applicable for the Employee Benefits Schemes. In view of this, it should be noted that focal point of answers is expected to be India Specific application. However, if application specific to any other country is quoted in the answer the candidate should answer the question with reference to Indian environment.*
- 3. Mark allocations are shown in brackets.*

- Q. 1)** An Indian PSU company, having 15000 employees, is offering various post-employment benefits like Gratuity, Provident Fund and Defined Contribution Pension Scheme to its employees. It is also covering medical benefits of the in-service employees through its Group Medical Insurance Policy. The cost of this medical insurance plan is shared between the employees and employer in the ratio of 50% (employees) and 50% (employer). The annual Medi-claim covers employee and his/her spouse. The medical benefit sum assured varies between INR 10 lacs to INR 50 lacs depending on the employee designation.

The average age of the employees is 30 years. With a view to encourage long term service, the company is planning to introduce Post-Retirement Medical Benefit (PRMS) to its in-service employees. The proposed benefits are similar to the in-service benefits offered under Group Medi-claim. It is also proposed that a PRMS fund is to be set up to which both the employer & the employee are expected to contribute equally (50:50).

The Company is engaging you as an actuarial consultant to offer advice on the management of the PRMS fund.

- i)** Describe the approach, you will use, to determine the annual contribution of the Company & the employee, to the fund. (10)

The PRMS Scheme has run for several years. The significant fund has built-up & there have been very few pay outs. There is a proposal to extend the medical benefits to those employees retired before the introduction of the Scheme & the company is actively considering the proposal. The profile of the retired employees is given as under:

Age group	Number	Age group	Number	Age group	Number
60-65	500	65-70	2000	>70	1500

- ii)** Discuss various factors that the management should consider while extending the medical benefits to these retired employees. (8)

Decision is now taken by the management to extend the benefits to this group. It is considering the following two alternatives for implementing the proposal.

- a) Include the retired employees under the existing (in service) Group Medi-claim policy.
b) Admit the retired employees into the PRMS fund.

- iii)** Discuss the advantages & disadvantages of these two alternatives taking into consideration the interests of different stake holders. (12)

The company, after considering the above alternatives, has adopted a hybrid approach where by retired members aged less than 70 will be covered as per alternative (a) & retired members aged >70 under alternative (b).

Three years after the implementation of the Scheme to the retired members, the company finds that there has been significant erosion in the PRMS fund & the annual contributions of active employees, retired members & the company has been increasing steadily. The company wants to understand the reasons for this adverse situation & the possible solutions to revert the trend.

- iv) Discuss the investigations to be carried out to find the possible reasons for such an experience and suggest the possible measures the company can take to control the costs. (12)
 - v) Discuss how the accounting (as per Ind AS19) of the medical benefits in the books of the company will differ under two different financing
 - a) PRMS fund approach
 - b) Group Medi-claim approach (8)
- [50]**

Q. 2) A large private sector Company, Energy Pvt. Ltd. (EPL) operating in the electric power distribution industry has acquired another public sector Company, XYZ in an asset sale agreement. XYZ offers a defined benefit Pension plan linked to final basic salary where pension amount would be calculated as $1/60 \times \text{final salary} \times \text{past service}$ whereas there is no such defined benefit pension plan offered by EPL. EPL offers an optional Defined Contribution NPS to their employees which is currently opted by less than 20% of their workforce. The Finance team, trustees and other stakeholders of EPL are not experienced with the administration, accounting and matters related to a Defined Benefit Pension plan. You have been the Actuary for EPL for the last 4 years and you are asked to help with the evaluation of overall feasibility and cost of the acquisition with respect to this pension plan.

- i) State the information that should be obtained for the overall due diligence and assessment of the cost of Defined Benefit Pension plan XYZ. (8)

It is noted that the scheme of XYZ was 80% funded on 31st March, 2021 and the next year's current service cost for the scheme is estimated to be 15% of the past service liability & EPL has decided to maintain the same funding level for the year 2021-2022 with assumptions of Discount rate of 6.5%; pension inflation rate of 4% and salary escalation rate of 7%.

- ii) Calculate the contribution as proportion of past service liability that the Company should make to (at least) maintain the funding level at the same level of 80% in the next one year. State the assumptions, if any, used in the calculations & also outline the factors to be considered by EPL while adopting this funding strategy. (8)

The scheme of XYZ provides an option to commute 1/3rd of the pension at retirement which then gets restored after 15 years if the member is alive. The value of commutation is calculated using the commutation table as applicable for the Government employees and at age 60 years the factor is 9.81 for INR 1 of pension p.a. The factors have not been revised for many years. A retiring member has asked for an explanation on how the commutation factors are computed and if the same should be revised in light of the changing economic and demographic patterns.

- iii) Explain what is commutation factor & state the points to be covered while addressing the queries of the retiring member. (8)

EPL has neither extended this Defined Benefit Pension scheme to its original employees nor will be offering it to any new entrants after the acquisition. EPL is now also considering to close the scheme to future accrual for the transferred in employees. The transferred in active members will retain linkage to final salary at retirement, however the service would be frozen. Also, the retirement age of these transferred in members will be revised from 58 years to 60 years to harmonise with the normal retirement age of EPL. However, if the members opt to retire early between ages 50 to 60, the pension benefit will start immediately and the pension amount would be reduced at 2% for every year the age falls short of 60 years.

- iv)** Discuss the implications of the above changes to the new employer EPL, its employees & to the members of the XYZ pension scheme. Also state possible ways to address any issues / risks arising due to these changes.

(10)

A transferred in employee (of XYZ) who is 56 years old and has a past service of 22 years is concerned about the above changes and has asked for an illustration of how the above changes will impact his pension amount if he opts for early retirement at 58 years as against retiring at the new normal retirement age of 60 years. His current monthly pensionable salary is INR 100,000.

- v)** Draft a response to the member with a suitable illustration on the changes in the pension due to early retirement. The draft should also state the factors to be considered by member while deciding to retire early.

(6)

Further it is noted that the pensioners and family pensioners are being paid directly by the sponsor. In order to reduce the risk and administration hassle, EPL is contemplating to buy annuities from an Insurance Company.

- vi)** Discuss the factors to be considered by EPL before deciding to buy annuities from insurers.

(5)

A new standard mortality rate table has been published for annuitants of insurance companies. The mortality rates of annuitants have reduced significantly under the new tables. The Company EPL would like to have an “impact analysis” of the new mortality table, if adopted, on the financials of the XYZ pension scheme.

- vii)** Describe how would this change impact the liabilities for Actives, Pensioners and Family Pensioners under both situations –

- a) sponsor continuing to pay pension
- b) sponsor deciding to buy out annuities from an insurance Company

Also explain how will the impact of this change in mortality rate be recognized under IND AS19.

(5)

[50]
